# Data Analytics (DA)

# Module-End -1

# Module End Assignment:

# “Data Cleaning, Analysis & Visualization with Excel”

# Assignment Task Overview

**Create and analyse an E-commerce dataset in Excel. The assignment should demonstrate the following:**

**Dataset Link: [E-commerce store dataset]**

- **Data entry, import, and basic organization using tables**
- **Data cleaning: find/remove duplicates, handle missing/inconsistent data, correct formatting errors, and validate data**
- **Remove the column that contains irrelevant data if it exists**
- **Application of Excel formulas for calculations, summaries, and logical/text operations**
- **Use the XLOOKUP/VLOOKUP functions to combine the tables.**
- **Use of Pivot Tables for advanced summarization**
- **Descriptive statistics (mean, median, mode, standard deviation, etc.) using formulas or Analysis ToolPak**
- **Data visualization: create at least three chart types, build an interactive dashboard, and use basic PivotCharts**

**Deliverables:**

- **Final Excel file with all steps visible (use multiple sheets or tabs)**
- **Screenshots illustrating each cleaning/processing step (optional but encouraged)**
- **One-page summary explaining key findings and decisions in word file**

# Skills to be Evaluated and Mapping

| Assignment Section | Core Skill | Sub-Skill | Expected Proficiency |
|---|---|---|---|
| Data Entry & Organization | Spreadsheet Mgmt | Table Creation, Import, Entry | Beginner |
| Data Cleaning & Transformation | Data Cleaning | Duplicates, Missing/Inconsistent Data | Intermediate |

| | | | |
|---|---|---|---|
| Excel Formulas & Functions | Statistical/Logic | SUM, IF, COUNTIF, TEXT, Average,xlookup/vlookup functions | Intermediate |
| Descriptive Statistics | Data Analysis | Mean, SD, Mode, Analysis ToolPak | Intermediate |
| Data Summarization (Pivot Table) | Data Summarization | Creating, Connecting Tables | Intermediate |
| Data Visualization | Visualization | Bar, Column, Pie, Sparklines, Dashboards | Intermediate |
| Reporting & Insights | Communication | Clear Findings Summary | Beginner |

## Rubric & Mark Allocation (Total 25 Marks)

| Skill/Component | Full Marks | Criteria for Full Marks | Partial Credit Description | Partial Credit Marks Range |
|---|---|---|---|---|
| Data Entry & Table Setup | 3 | Clean tables; correct import; data types properly formatted | Basic setup, but some improper import/types or minor errors | 1 – 2 |
| Data Cleaning & Transformation | 5 | All cleaning steps correct; missing/inconsistent fixed, combining the tables if necessary; and it is well-documented | Most steps correct; minor errors or incomplete documentation | 2 – 4 |

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|---|---|---|---|---|
| Excel Formulas & Functions | 4 | Correct use of SUM, IF, COUNTIF, TEXT; comprehensive application | Some formulas applied, but inconsistent/partial | 2 – 3 |
| Descriptive Statistics | 4 | Accurate calculations; used Analysis ToolPak/similar tool properly | Most stats correct; process partially shown | 2 – 3 |
| Data Summarization (Pivot) | 3 | Created/used Pivot Table to summarize multi-dimensional data | Some fields summarized; minor or moderate issues | 1 – 2 |
| Data Visualization | 4 | ≥3 chart types; clear labels; dashboards interactive/organized | Charts missing/labels unclear/dashboards basic | 1 – 2 |
| Reporting & Insights | 2 | Clear, concise one-page summary linking findings to visuals | Partial summary, omits some findings or interpretations | 1 – 1 |

# DA Skill Taxonomy Structure

- **Spreadsheet Management**
  - **Data Entry, Import, Table Creation**
- **Data Cleaning**
  - **Duplicates Removal, Missing Data Handling, Consistency Checks**
- **Formulas & Analysis**
  - **SUM, IF, COUNTIF, AVERAGE, TEXT Functions, Statistical Measures**
- **Data Summarization**
  - **Pivot Tables, Filtering, Sorting**
- **Visualization**

    - **Charts, Dashboards, PivotCharts**
- **Communication**
    - **Summary, Reporting Key Insights**

## Evaluation Guidelines

1. **Open and review all Excel sheets/tabs provided.**
2. **For each section, match student work to the specific skill and sub-skill area in the taxonomy.**
3. **Assign full or partial marks per the rubric, using suggested partial mark ranges for "partially complete" work.**
4. **For ambiguous or incomplete submissions, note what's missing and correspondingly reduce credit.**
5. **Comments must explain strengths and improvement points for each section.**
6. **Objective assessment: focus on function, correctness, clarity, not on visual design beyond the stated requirements.**

## Achievement Criteria & Passing Rules

- **At least partial (not zero) credit must be received in every section—no component can be left unattempted.**
- **Students must fully achieve at least two major skills: Data Cleaning & Transformation and Data Visualization.**
- **Feedback and a detailed scorecard must be given for all evaluated assignments.**